

Innovation

BACKTRACK: Your GPS for Learning
University of the Philippines Manila
234 words · Maximum 300 words

BACKTRACK's organizing question is: what is blocking the thing this learner needs now, and which useful step could get them back? The destination remains visible while the route changes with their answers.

ALEKS and modern AI learning tools already offer adaptive practice, prerequisite checks, and goal-based study plans. BACKTRACK concentrates those ideas into a recovery routine around Khan: a teacher-confirmed destination, a visible route, relevant practice, fresh evidence of return, and a reusable school kit.

Two wrong answers can need different help. A learner who finds the right factors but writes the wrong solutions can get a different first check from one who mixes up the factor pair. BACKTRACK explains that turn and tests it with fresh problems.

The reward is concrete. When a learner demonstrates a step, its review leaves the route. When a learner needs help, a deeper step can appear. Confidence changes support and follow-up checks, not mathematical correctness. The current product requires two distinct unassisted checks for a route-specific decision with later checks to see what stays with the learner.

Learners choose a manageable block, can stop without losing progress, and may take one optional next turn. A comeback check preserves history while reassessing what is useful now. We will measure voluntary continuation, frustration, and return visits.

The working website lets judges experience those decisions. The pilot will test whether this combination adds value beyond an equal-time teacher-selected Khan routine.